ANNA JOHNSON FLOORING — CHEN E-R DIAGRAM

request_area is a SQL VIEW, not a table.
It derives total_sqft and room_count from request_room.

BILL MESSAGE
sender
bill_id (FK)
message_id
msg_type
message
created_at

CLIENT
last_name
first_name
client_id
address
phone
email (UNIQUE)
card_number
card_holder
password_hash
card_exp_month
card_exp_year
registered_at

Submits

FLOORING REQUEST
service_address
client_id (FK)
request_id
pref_start_date
pref_end_date
budget
created_at
status
note
total_sqft
room_count

Has Photo

REQUEST PHOTO
photo_no
request_id (FK)
photo_id
file_name
uploaded_at

Discusses

Places

Becomes

About

Quoted In

BILL
created_at
version_no
order_id (FK)
bill_id
amount_due
note
status

Billed By

SERVICE ORDER
client_id (FK)
quote_id (FK, UNIQUE)
order_id
service_address
agreed_price
scheduled_start
created_at
status
scheduled_end
actual_install_date
completion_date
completion_note

Contains

NEGOTIATION MESSAGE
sender
quote_id (FK, nullable)
request_id (FK)
message_id
msg_type
message
created_at

Answers

QUOTE
created_at
version_no
request_id (FK)
quote_id
waste_pct
material_cost
labor_cost
proposed_end
proposed_start
total_price
other_cost
note
status
decided_at

Paid By

Includes

Itemizes

Lists

PAYMENT
amount
bill_id (FK, UNIQUE)
payment_id
paid_at
card_last4

ORDER ITEM
order_id (FK)
order_item_id
room_id (FK)
product_id (FK)
est_qty_sqft
installed_sqft
actual_qty_sqft
unit_price_sqft

Installs

REQUEST ROOM
room_name
request_id (FK)
room_id
room_type
length_ft
width_ft
current_flooring
product_id (FK)
sqft
removal_required
install_pattern

For Room

QUOTE ITEM
item_type
quote_id (FK)
quote_item_id
room_id (FK, nullable)
product_id (FK, nullable)
description
unit
quantity
area_sqft
unit_price
amount

Wants

Uses

Of Product

BILL ITEM
item_type
bill_id (FK)
bill_item_id
description
amount

PRODUCT CATEGORY
category_id
category_name (UNIQUE)

Belongs To

PRODUCT
brand
product_name
category_id (FK)
product_id
material_style
color
unit_price
width_in
length_in
pricing_unit
sqft_per_unit
description
is_active

Contractor is an independent login entity.
No foreign-key relationships connect it to other tables.

CONTRACTOR
username (UNIQUE)
contractor_id
password_hash
full_name

CHEN NOTATION
Rectangle = Entity
Diamond = Relationship
Oval = Attribute
Underlined = Primary key
Dashed oval = Derived attribute
Thin line = Partial participation
Thick line = Total participation
Thin arrow = At most one
Thick arrow = Exactly one

CONNECTIVITY COLORS
Blue = Client / Request
Teal = Product / Room
Purple = Quote workflow
Pink = Messages
Orange = Orders / Installation
Green = Billing / Payment